UNIT-2: Multiple Integrals
Appearance Year: 2025-26

1. Beta and Gamma Functions:
   - Evaluate $\int_0^2 x(8 - x^3)^{1/3}\, dx$.
2. Change of Order of Integration:
   - Change order & evaluate $\int_0^a \int_{ax}^{a} \frac{y^2\, dy\, dx}{\sqrt{y^4 - a^2 x^2}}$.
3. Applications to Area & Volume:
   - Prove area in positive quadrant bounded by $y^2=4ax$, $y^2=4bx$, $xy=c^2$, $xy=d^2$ is $\frac{1}{3}(d^2-c^2)\log(b/a)$.
   - Find volume bounded by coordinate planes and $\sqrt{x/a} + \sqrt{y/b} + \sqrt{z/c} = 1$.